Filament Dryer



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Filament dryer

What it is used for

A filament dryer removes moisture from moisture-sensitive 3D-printing filament, such as nylon, PETG, or other compatible polymers, before use. Correct temperature, drying time, and spool handling help improve print consistency while preventing filament damage and overheating.

Main parts

- Insulated drying chamber
- Heating element and temperature controller
- Air circulation fan or airflow system
- Temperature sensor and display
- Spool holder or filament support
- Power cable, door or lid, and safety controls

Before you start

1. Confirm the filament material, manufacturer drying recommendation, spool type, and required drying duration.
2. Check that the dryer is suitable for the filament and that its temperature range will not damage the spool or material.
3. Inspect the chamber, heater, fan, sensor, controls, door seal, and power cable for damage.
4. Ensure the chamber is clean and free of incompatible filament fragments or foreign material.
5. Verify that the temperature display and safety cut-outs function correctly.
6. Place the dryer on a stable, heat-resistant surface with adequate clearance around ventilation openings.

How to use

1. Switch off the dryer before loading or unloading a spool or adjusting the spool holder.
2. Prepare the filament spool according to the manufacturer's recommendations and confirm that it can tolerate the selected drying temperature.
3. Place the spool securely in the chamber and arrange the filament path to prevent tangling or contact with hot components.
4. Close the door or lid and ensure that ventilation openings remain unobstructed.
5. Switch on the dryer and set the temperature and duration recommended for the filament material.
6. Allow the chamber to reach the set temperature and verify the displayed reading before relying on the drying cycle.

7. Monitor the temperature and dryer status throughout the process, especially during the initial heating period.
8. Stop the dryer if the temperature rises unexpectedly, the fan fails, smoke appears, or an unusual smell develops.
9. Do not exceed the filament or spool manufacturer's temperature limits; reduce the setting if the spool begins to deform.
10. At the end of the cycle, switch off the heater and allow the spool to cool according to the material instructions before handling.
11. Remove the spool carefully and keep the dried filament in a sealed, moisture-resistant container or suitable dry storage.
12. Feed the filament into the 3D printer according to the printer and material instructions.
13. Switch off and unplug the dryer before cleaning or performing maintenance.

⚠ Safety

- Use the dryer only for compatible filament materials and within the manufacturer's temperature limits.
- Do not leave the unit operating if the manufacturer prohibits unattended operation; follow the specified supervision requirements.
- Keep the dryer away from water, flammable materials, and blocked ventilation areas.
- Do not touch the heating element or internal surfaces while hot; allow adequate cooling time.
- Never cover the ventilation openings or modify the thermostat and safety cut-outs.
- Disconnect power before cleaning, opening service panels, or inspecting electrical components.
- Stop using the dryer if there is smoke, burning smell, damaged wiring, or abnormal heating.

Care & storage

- Allow the dryer to cool and disconnect power before cleaning the chamber.
- Remove filament fragments and dust using a method approved by the manufacturer.
- Inspect the fan, temperature sensor, door seal, and power cable regularly.
- Keep air passages clear and check that the spool holder rotates freely where applicable.
- Arrange servicing if temperature control becomes inaccurate, airflow weakens, or the dryer overheats.

Common mistakes to avoid

- Using the same drying temperature for every filament type.
- Exceeding the safe temperature of the filament or its spool.
- Blocking airflow or allowing the filament to tangle during drying.
- Leaving dried filament exposed to humid air instead of storing it in a sealed container.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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